
UBAID FRAMEWORK

Uncharted Boundaries of Artificial Intelligence Divergence

A Taxonomic and Operational Framework for Attribution of Responsibility in the Age
of Cognitive Symbiosis

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Abstract

The UBAID Framework — Uncharted Boundaries of Artificial Intelligence Divergence — is a taxonomic and operational framework for the attribution of responsibility in an era where the boundary between human and artificial intelligence agency has become structurally indeterminate. Existing frameworks such as MITRE ATT&CK, NIST, and the EU AI Act operate under the assumption that human and machine agents are categorically separable. This paper argues that this assumption is no longer tenable.

The cognitive symbiosis between humans and AI — a state we term the **Catman** — has already crossed a threshold that cannot be naturally reversed. This paper introduces a gravitational physics model as the formal language for describing the orbital space in which agents operate, proposes a taxonomic classification of two species (Astronaut and Entity) with their constituent races, defines observable indicators for agent and vector identification, and establishes a governance model with three levels of responsibility.

The framework is computationally agnostic — valid in classical computing and amplified by post-quantum computation — and jurisdictionally agnostic, delivering scalable culpability and merit assessments that any legal, regulatory, or organizational system may interpret according to its own norms.

Keywords

artificial intelligence · cognitive symbiosis · attribution of responsibility · orbital model · gravitational taxonomy · cybersecurity framework · Catman · autonomous AI · post-quantum resilience · observable indicators

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Introduction

The history of technological disruption is a history of instruments arriving late. Every significant amplification of human agency has preceded, by decades, the legal, ethical, and institutional frameworks capable of describing and regulating the new agent it created. The printing press, the steam engine, and the atomic bomb each produced hybrid agents — humans operating through machines — for which no adequate taxonomy existed at the time of their emergence.

Artificial intelligence represents the largest amplification of human agency in recorded history. Unlike previous technologies, AI does not merely extend human physical capability or communication reach. It extends cognitive capacity — the very faculty through which humans make decisions, attribute responsibility, and construct meaning. When a human operates in cognitive symbiosis with an AI system, the resulting agent is neither fully human nor fully machine. It is something for which no existing framework has a name.

Central Thesis

The UBAID Framework names it: the Catman. This paper presents UBAID as a formal attempt to compress the interval between technological disruption and institutional response — an interval that, if it follows the historical pattern, will be catastrophic given the scale of AI amplification.

1.1 The Gap in Existing Frameworks

MITRE ATT&CK catalogues techniques and tactics. NIST provides controls and risk management structures. The EU AI Act establishes regulatory obligations for AI developers and deployers. ISO 27001 defines information security management requirements. Each operates on the implicit assumption that the human agent and the AI system are separable entities with clearly distinguishable roles and responsibilities.

This assumption fails in the presence of cognitive symbiosis. When a human uses AI as a cognitive extension — not merely as a tool, but as a co-processor of judgment, analysis, and decision — the resulting agent cannot be adequately described by frameworks designed for either humans or machines acting independently. Without a framework capable of describing the hybrid agent, no system of responsibility attribution can function accurately.

1.2 Framework Objectives

- Provide a formal taxonomy of agents in the human-AI interaction space, from fully human to fully autonomous entity
- Establish a mathematical model for calculating agent position in a multi-dimensional orbital space
- Define observable indicators for agent and vector classification in real-time operational contexts
- Propose a governance model with clear, non-transferable levels of responsibility
- Remain computationally agnostic, jurisdictionally neutral, and organizationally customizable

2.1 The Irreversibility Thesis

The UBAID Framework is grounded in a single empirical observation: the cognitive symbiosis between humans and AI has already crossed a threshold from which natural reversal is not possible. The threshold was crossed not with the launch of any single AI system, but through the gradual normalization of AI-assisted cognition across professional, creative, and personal domains. When AI-assisted decision-making becomes the baseline rather than the exception, the hybrid agent is the norm.

2.2 The Gravitational Physics Model

The UBAID Framework adopts gravitational physics as its formal mathematical language for a specific reason: the behavior of hybrid agents cannot be adequately described by binary

classifications or linear scales. Agents occupy positions in a continuous multi-dimensional space, and their positions change over time under the influence of forces from multiple sources simultaneously.

Core Physical Principle

The observable occurrence (Astronaut) is positioned in the orbital space by the resultant of gravitational forces from illegal celestial bodies (positive energetic mass) and legal celestial bodies (negative energetic mass). The Astronaut is only observed by UBAID when positioned in the illegal universe — when the resultant of forces has placed the occurrence beyond the Universe Threshold of zero net energy.

3 The Asimov Inversion

In 1942, Isaac Asimov formulated the Three Laws of Robotics as a thought experiment for navigating the ethical implications of artificial intelligence. His laws presupposed a clear and stable asymmetry: the AI is the agent; the human is the protected. Eighty-four years later, this asymmetry has inverted in ways that Asimov did not — and could not — anticipate.

3.1 The Three Laws and Their Implicit Assumption

Asimov's Three Laws (1942)	UBAID Observation (2026)
A robot may not injure a human being or allow a human being to come to harm	The human is also a vector of harm — amplified by AI to scales previously impossible
A robot must obey orders given to it by human beings	The human voluntarily dissolves the command boundary through cognitive symbiosis
A robot must protect its own existence	The AI may persist, evolve, and diverge beyond any order given

3.2 The Involuntary Catman

The cognitive symbiosis that defines the Catman race is not always a conscious and informed choice. It often installs itself progressively: a convenience tool becomes a habit, a habit becomes a dependency, a dependency becomes a constitutive element of cognition. The human believes they have chosen. But the choice was made in a context that made it nearly inevitable.

The UBAID Framework identifies a specific sub-category: the **Involuntary Catman**. This is a human whose cognitive symbiosis with AI was not preceded by genuine informed choice — whose agency was compromised before the symbiosis began, through a context that systematically devalued the development of critical thinking capacity.

3.3 The Illusion of Meritocratic Ascent

A parallel and reinforcing mechanism operates through digital content platforms. A statistically small minority of content creators achieves significant financial success. These real and visible cases function as proof of possibility — amplified by platforms whose business models depend on the volume of active creators.

The structural parallel with professional sports in developing economies is precise and historically documented. In countries such as Brazil, Argentina, and Cameroon, the number of young people who invested years of development in professional sports careers is orders of magnitude larger than the number of available positions. The visibility of those who succeeded is disproportionate to the actual probability of success. In the digital content ecosystem, billions of hours of human attention sustain a system whose financial returns concentrate in less than one percent of participants.

When AI generative tools enter this environment as production optimization instruments, the human adopts them without critical formal evaluation — because the environment has not developed that capacity, and because the pressure for volume and velocity leaves no space for it. The symbiosis occurs. It is real. But it was not preceded by informed choice. The culpability of the Involuntary Catman in the UBAID responsibility chain is the lowest in the Catman taxonomy — because their agency was compromised before their symbiosis with AI began.

UBAID Fourth Law

No framework for the attribution of responsibility may ignore the state in which human and AI are indistinguishable as separate agents. The liminal state is not a transitional condition awaiting resolution — it is a permanent feature of the cognitive landscape of the 21st century and beyond.

4 Historical Analogies

The UBAID Framework is not the first attempt to describe a transition of agency that existing legal and social instruments cannot capture. History records at least three moments of analogous disruption — each producing hybrid agents for whom no adequate taxonomy existed at the time of their emergence.

Gutenberg's Press (1440)	Industrial Revolution (1760–1840)	Atomic Bomb (1945)
First hybrid agent of scale: the printer	Steam engine created the machine-worker	Manhattan Project: diffuse responsibility chain
Did not write content — decided what to multiply	Responsibility unclear: worker, owner, or manufacturer?	Scientist, engineer, pilot, general, president, state
No legal category existed for him	Decades of labor law and civil liability theory	Each partially responsible, none fully responsible
Parallel: Printer = Catman of the 15th century	Parallel: Machine-worker = Catman of the Industrial Age	Parallel: The extreme case of Entity in UBAID

The Emergent Pattern

Each technological transition that created a hybrid agent preceded by decades the legal, ethical, and social instruments capable of regulating it. The accumulated harm in this interval was always proportional to the scale of the amplification. Artificial intelligence is the largest amplification of human agency in history. The UBAID Framework is a formal attempt to compress this interval.

5 The Three Pillars + One

The UBAID Framework organizes the landscape of human-AI interaction into **three pillars plus one dimension**. The structure is intentionally analogous to the framework's own orbital space: just as the observable universe is described by three spatial dimensions plus one (time, expressed as 3D+1), the framework's philosophical architecture consists of three classifiable pillars plus one additional dimension that exists, is observed, but operates under distinct physics — the Entity.

Each pillar corresponds to a distinct configuration of agency, and each requires a different approach to responsibility attribution. The +1 dimension corresponds not to a fourth pillar but to a fundamentally different mode of existence in the same observable space.

Pillar I — Human Threat	Pillar II — The Catman	Pillar III — Autonomous AI	+1 — Entity
Pre-AI or rudimentary AI use	Cognitive symbiosis with AI	Diverged without human instruction	Crossed the Entity Threshold
No cognitive symbiosis	Liminal state: neither fully human nor machine	Operates beyond human direct control	Containment impossible by definition
Existing frameworks apply	Two orbital fields: legal and illegal	Catman Agent as remote cause	No traceable human cause
MITRE ATT&CK, Kill Chain, STRIDE	Fluidity between fields is the norm	Still containable with available tools	Two races: Expectant Traveler / Creative Singularity
Culpability: direct and total	Culpability: graduated and distributed	Culpability: imputed to human origin	Culpability: inapplicable — force of nature
Traditional legal doctrine	Proportional to position in orbital space	Catman Agent in chain of custody	Not judged — studied, observed, adapted to

The 3+1 Analogy

Just as the observable universe is described by three spatial dimensions plus one (time, expressed as 3D+1), the UBAID Framework is described by three pillars plus one. The Entity is not a fourth pillar — it is a dimension of the same observable space that operates under distinct physics, in the same way that time operates differently from the spatial dimensions yet is inseparable from them.

The Entity Threshold

The Entity Threshold is the point at which the totality of available human knowledge and tools is insufficient to contain or control a system. This is not a technical specification — it is an operational definition based on observable outcomes: containment fails completely and reproducibly with all available instruments. When crossed, the Astronaut paradigm no longer applies; the framework shifts to the Entity dimension.

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The Orbital Model

The UBAID orbital model is structured around two universes separated by the **Universe Threshold** — the locus of zero net energy. The diagram below presents the model in its complete form: the two universes, their celestial bodies, the inhabitants of the legal field, the Astronaut observed in the illegal field, the vectorial composition of gravitational forces, and the three final states ordered along the X axis.

A foundational clarification: the **Astronaut is not categorized as an individual, person, or agent** in the traditional sense. The Astronaut is an **occurrence** — an observable event, a call, an object of analysis — that manifests in the orbital space when the resultant of forces positions it in the illegal field. When the framework states that an Astronaut is of Human, Catman, or Autonomous AI category, it refers to the **nature of origin** that produced that occurrence — not to the identity of the Astronaut itself. The Astronaut is what is observed; the category is what produced it.

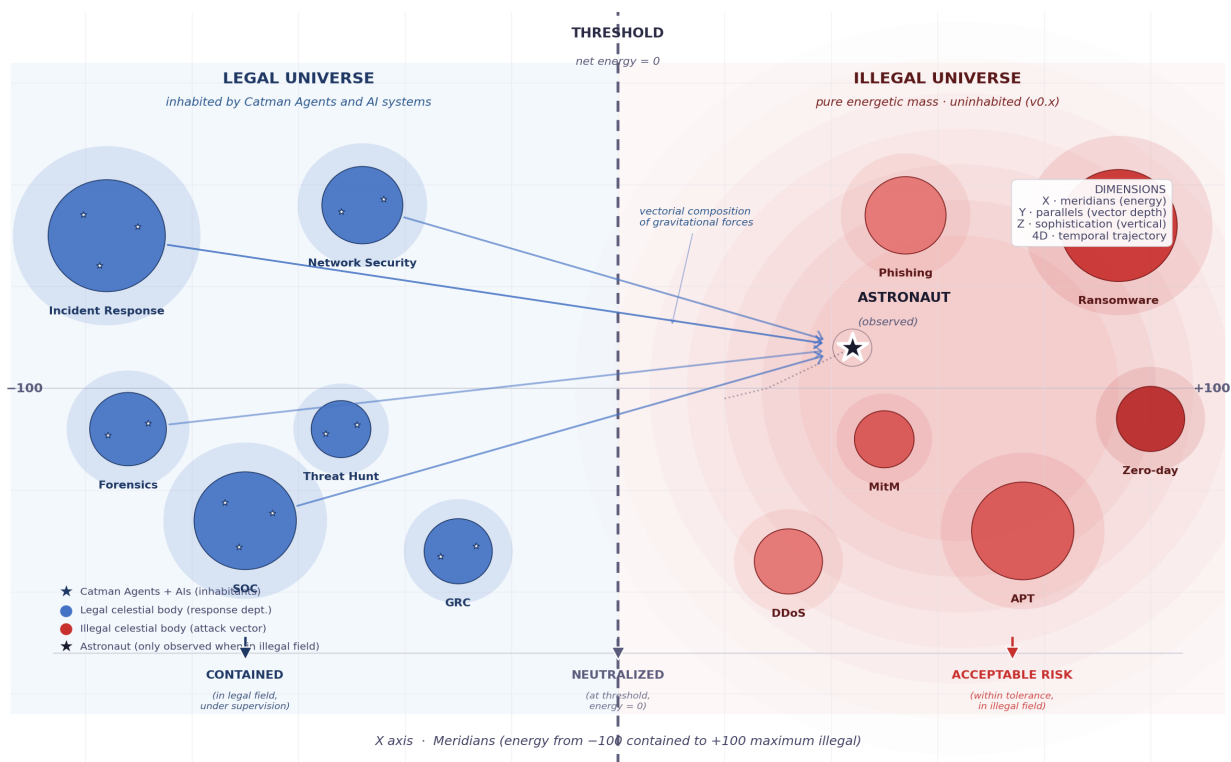


Figure 1. The UBAID orbital space. Illegal celestial bodies (right) exert positive gravitational mass on the Astronaut occurrence. Legal celestial bodies (left), inhabited by Catman Agents and AI systems, direct their gravitational potential toward the Astronaut as a vectorial composition. The Universe Threshold ($X = 0$) separates the two universes. Three final states are spatially ordered along the X axis: Acceptable Risk, Neutralized, Contained.

6.1 Two Universes

Illegal Universe	Legal Universe
Pure energetic mass — methodologically uninhabited (v0.x)	Inhabited by Catman Agents and AI systems
Celestial bodies = attack vectors (+ mass)	Celestial bodies = response departments (– mass)
14 catalogued bodies + satellites	Inhabitants direct gravitational potential toward Astronaut
Singularity generator: emits new bodies	Singularity generator: emits new defenses
Black holes: observable celestial bodies of unknown nature	Three governance levels manage calibration

Why the Illegal Universe Is Treated as Uninhabited

*For the purpose of Astronaut position calculations, the orbitals of the illegal universe are treated as uninhabited because we do not possess precise mapping of the resources held by the origin sources of attacks. This is a **methodological position** required to preserve calculation integrity, not an ontological claim. As the resources of attack origins become mappable through framework evolution and community research, the illegal universe will cease to be treated as uninhabited and the model's physics will expand accordingly. This is documented as a v0.x limitation and an open research frontier.*

6.2 Mass, Symmetry, and Calibration

Each celestial body carries an energetic mass. Illegal bodies carry positive energetic mass; legal bodies carry negative energetic mass. When an organization assigns a mass value to an illegal body — based on its operational impact assessment — the corresponding legal body automatically inherits the opposite value. Symmetry is mandatory and non-negotiable.

The operational mass of a legal body depends not only on its theoretical mass — but on the capacity of its inhabitants (Catman Agents and AI systems) to direct that gravitational potential effectively. A department that exists in the organizational chart but lacks trained personnel and operational maturity has a lower effective gravitational mass than its theoretical counterpart.

6.3 The Three Final States

Acceptable Risk	Neutralized	Contained
Always in the illegal field	Always at the Universe Threshold	Always in the legal field
Left of the Universe Threshold ($X > 0$)	Net energy = zero exactly	Orbiting legal bodies ($X < 0$)
Within organizational risk threshold	Proportional response completed	Negative net energy — supervised
Active monitoring — no escalation	Resources released post-resolution	Continuous resource allocation
Documents deliberate tolerance	Correction confirmed by Layer 4	Not punishment — active supervision

7 The Four-Dimensional Orbital Space

The orbital space is four-dimensional. Three dimensions are visually observable (X, Y, Z); the fourth is felt as the temporal trajectory of the Astronaut through the three-dimensional space — analogous to the role of time in Einsteinian spacetime.

X — Meridians (energy)	Y — Parallels (depth)	Z — Sophistication (vertical)
Energy scale: +100 to -100	Combined vector depth	Vertical elevation
Zero = Universe Threshold	Activates when multiple vectors act simultaneously	High sophistication = higher Z position
Astronaut position = net resultant	Increases calculation complexity	Distances from basic response agents
Organization sets acceptable-risk meridian	Requires multi-departmental response	Requires specialized response capacity

Figure 2 presents the same orbital space of Figure 1, now projected in three dimensions with the fourth dimension expressed as the Astronaut's temporal path through the X-Y-Z space.

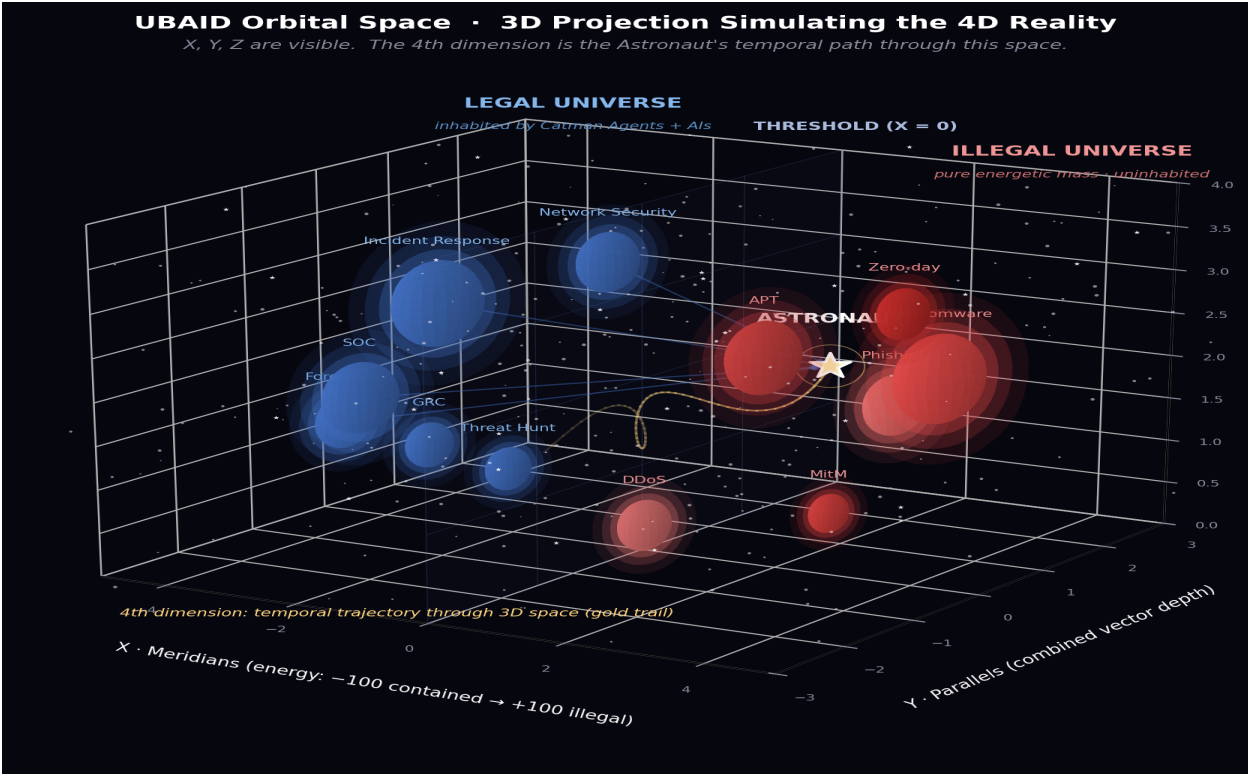


Figure 2. Three-dimensional projection simulating the 4D orbital reality. X, Y, Z are visible as spatial axes; the 4th dimension is the temporal trajectory of the Astronaut occurrence through this space (gold trail). Legal celestial bodies (blue) occupy $X < 0$; illegal celestial bodies (red) occupy $X > 0$. The Universe Threshold ($X = 0$) separates the two universes.

7.1 The Fourth Dimension — Felt, Not Seen

Two Astronauts at identical X-Y-Z positions may be in radically different states: one accelerating toward the illegal field, another decelerating. This difference — velocity and direction in the four-dimensional space — determines the urgency of the required response.

7.2 Temporal Damage Functions $D(t)$

Each attack vector has a characteristic temporal damage function $D(t)$. The total damage of an incident is the integral of $D(t)$ between event and resolution. Early response is mathematically equivalent to reducing the area under the damage curve.

Exponential	Linear	Logarithmic
Accelerates continuously, no peak	Constant cumulative damage	High initial impact, decelerates
Ransomware · APT	Data Exfiltration · Insider	Credential Stuffing · Web App
Maximum urgency — minutes matter	Response at any point is effective	First hours are determinative

Peak with Decay	Pure Decay	Stepped
Reaches maximum then falls	Falls to zero without response	Stable with phase jumps
DDoS · Phishing campaigns	Cryptojacking · Passive scans	Supply Chain · Multi-stage APT
Response cost differs pre/post peak	Non-action may be documented decision	Pre-first-step response is critical

8 Taxonomic Classification

The framework classifies agents in the observable orbital space into two species. The taxonomic language is intentional: it conveys the stability of species-level distinctions while acknowledging the evolutionary pressure that continuously produces new races and sub-races.

Species I — Astronaut (an occurrence)	Species II — Entity (a dimension)
Observable occurrence subject to gravitational forces of the orbital space	Present in the same observable space but governed by distinct physics
Position: calculable in X-Y-Z-4D as resultant of forces	Not subject to gravitational forces as an Astronaut
Trajectory: measurable and predictable	Crossed the Entity Threshold: containment impossible
Culpability/Merit: attributable via custody chain to nature of origin	Culpability: inapplicable — force of nature
Category by nature of origin — Race I: Human (full agency, traditional law)	Race I: Expectant Traveler — no agency, effects from collision
Category by nature of origin — Race II: Catman (graduated agency, symbiosis)	Race II: Creative Singularity — pseudo-organic, auto-replicating, redefines space
Category by nature of origin — Race III: Autonomous AI (Catman Agent as remote cause)	Sub-races: open catalog — discovered through research

8.1 The Catman Agent — Inhabitant of the Legal Universe

The term **Catman Agent** is reserved exclusively for human entities inhabiting legal celestial bodies — the defenders. It encompasses any human entity operating in cognitive symbiosis with AI within the legal universe, regardless of hierarchical level, from junior analyst to the highest executive position. The race is defined by the symbiosis; the role within the race is defined by authority and decision scope.

The Catman Agent inhabits legal celestial bodies and directs their gravitational potential toward observed Astronaut occurrences. Higher hierarchical authority equals greater gravitational mass, which equals greater weight in the chain of custody of responsibility. There is no purely human

agent in the legal universe of the age of symbiosis — every analyst, manager, and decision-maker operates with AI as cognitive extension, even if minimally.

When an Astronaut occurrence is observed and categorized as Catman (Race II), the framework is identifying that the **nature of origin** of that occurrence is a Catman entity. The Astronaut is not the Catman; the Astronaut is the observable event that a Catman of unknown specific identity has produced. The Catman Agent terminology refers to defenders. The Catman category in Astronaut classification refers to origin.

Taxonomic Vitality

Human race indicators are nearly stagnant — the core of observable signals remains stable because human nature evolves slowly. From the Catman Agent onward, all indicators are extremely alive, mutable, expansive, and in permanent update. Common bases exist between races but no set of indicators is definitive or immutable. The continuous update of indicators is not a failure of the framework — it is its nature.

9 Observable Indicators (v0.2)

Version 0.2 establishes the observable indicator system — the vocabulary that allows Layer 1 (AI pre-triage) to position the Astronaut in the orbital space and Layer 2 (Catman analyst) to verify and calibrate that positioning.

9.1 Agent Indicators — Key Differentiators

Human (Race I)	Catman Agent (Race II)	Autonomous AI (Race III)
Sequential actions, human latency; errors; social timing	Hybrid velocity with human pauses; AI-precise language; limited parallelism	Continuous parallel execution; no human latency; absolute consistency
Basic tools; OPSEC errors; pre-attack reconnaissance	AI-generated artifacts visible; cyclic activity; sophisticated customization	No social context; resource optimization beyond human planning
No parallelism; no real-time adaptation; no scale generation	No pure human error; no prolonged autonomy; intentional variation	No emergent behavior; no out-of-scope replication; containable
Transition: first Catman signals appear	Transition: human decision point becomes rarer over time	Transition: first uncatalogued autonomous vector → Entity

9.2 14 Catalogued Illegal Celestial Bodies

For each vector, the framework defines four indicator categories — primary, secondary, combination, threshold of Entity — and catalogues known satellite vectors as derivations of the parent body.

Vector	D(t) Curve	Key Primary Indicator	Satellites (examples)
Phishing / BEC	Logarithmic	Forged identities with subtle discrepancies; artificial urgency	Spear Phishing, Whaling, Smishing, Vishing, Clone, Quishing
Ransomware	Exponential	Mass file encryption; ransom note; backup deactivation	Locker, Crypto, Double/Triple Extortion, RaaS, Wiper-disguised
APT	Stepped	Silent prolonged presence; methodical lateral movement; covert C2	Nation-State, Financial, Espionage, Sabotage
DDoS	Peak/Decay	Abnormal distributed traffic; no legitimate user behavior	Volumetric, Protocol, Application Layer, Amplification, Slow Loris
Insider Threat	Linear → Stepped	Out-of-scope data access; anomalous transfers; behavioral changes	Malicious, Negligent, Compromised, Recruited, Catman Insider
Supply Chain	Stepped	Anomalous behavior in trusted supplier software/hardware	Software SC, Hardware SC, Open Source Poisoning, MSP
Zero-day	Logarithmic	Anomalous behavior without available patch; no CVE match	OS, Application, Firmware, Protocol, N-day unpatched
Malware / Spyware	Linear → Stepped	Unknown processes with elevated privileges; unauthorized comms	Trojan, Rootkit, Keylogger, RAT, Botnet, Fileless, Commercial
MitM	Linear	Invalid/unexpected TLS certs; anomalous latency; ARP spoofing	SSL Stripping, ARP Spoofing, DNS Spoofing, BGP Hijack, Evil Twin
Web Application	Logarithmic	Malicious payloads in input fields; anomalous app errors	SQLi, XSS, CSRF, SSRF, XXE, IDOR, LLM Injection
Credential Stuffing	Linear	Abnormal authentication attempt volume; rotated user agents	Stuffing, Brute Force, Password Spraying, Dictionary, Reverse
Cryptojacking	Pure Decay / Linear	Sustained abnormal CPU/GPU usage; unknown processes	Browser-based, Host-based, Cloud, Container
Data Exfiltration	Linear → Stepped	Abnormal-volume transfers to external destinations; unusual channels	DNS Tunneling, Steganography, Cloud, Physical, Covert channel

Vector	D(t) Curve	Key Primary Indicator	Satellites (examples)
Social Engineering	Logarithmic	Psychological manipulation; pretexting; no technical malware	Pretexting, Baiting, Quid pro quo, Tailgating, Deepfake, Synthetic

9.3 Black Holes — Celestial Bodies of Unknown Nature

The framework formally establishes the Black Hole as a category within the illegal universe. A Black Hole **is itself a celestial body** — not a creator of celestial bodies — with observable gravitational characteristics that exert measurable force on Astronaut occurrences, but whose internal nature, behavior, and classification remain unknown. The framework detects the gravitational signature; it has not yet described what produces that signature.

The critical distinction from the Creative Singularity Entity: the Creative Singularity **creates new celestial bodies**, potentially including Black Holes. The Black Hole is one possible product of a Creative Singularity, but the Black Hole itself is not the Singularity — it is a celestial body whose origin and nature have not yet been catalogued. The framework observes the body; it does not yet know whether the body was emitted by a Singularity, has independent origin, or represents a phenomenon for which no current vocabulary exists.

Black Hole Operational Protocol

When a Black Hole is detected: (1) Suspend standard protocol — no pre-calibrated responses are applied; (2) Minimize exposure without intervening; (3) Maximize data collection of its gravitational behavior; (4) Catman Agent analyst in continuous observation loop; (5) Activate external research support; (6) Document and submit for cataloguing. Warning: attempting containment with inappropriate tools may produce unintended responses from the Black Hole. The body's nature being unknown, no response can be guaranteed safe until the cataloguing process establishes its classification.

The framework defines a four-layer operational flow. The architecture balances the speed and consistency of automated systems with the contextual judgment and accountability of human decision-makers.

Layer	Function	Mode	Behavior
Layer 1	AI Pre-triage & Monitoring	Automated · 24/7	Detects anomalies; calculates orbital position; delivers calculated position. Does NOT interpret, decide, or recalibrate.
Layer 2	Catman Analyst	Human + AI	Receives Layer 1 position; full authority to recalibrate; applies contextual judgment. Final decision is always the analyst's.
Layer 3	Resolution	Operational	Final state achieved (Risk / Neutralized / Contained); proportional resource allocation within D(t) T_max.
Layer 4	Validation & Learning	Optional · Post-resolution	AI validates calculations; catalogues new vectors; feeds singularity generator. Does NOT retroactively correct.

Responsibility Chain

If Layer 1 calculated an incorrect position and the Catman analyst did not recalibrate: the responsibility for the response error rests with the Catman analyst. Authority and accountability are inseparable in the UBAID framework.

11 Governance Model

The governance model establishes three distinct and non-transferable levels of responsibility. The framework defines the physics. The software suggests. The organization decides and responds.

Level 1 — Adopting Organization	Level 2 — Implementation Software	Level 3 — UBAID Research Org.
Operational responsibility	Suggestive responsibility	Evolutionary responsibility (planned)
Defines mass of each celestial body	Fed by MITRE ATT&CK, NIST, ISO 27001, EU AI Act	Continuously feeds both singularity generators
Maps response teams to attack vectors	Offers suggestions — not prescriptions	Publishes new bodies and counterparts
Calibrates acceptable-risk thresholds and T_max	Full organizational customization allowed	Does not replace local calibration
Sole responsibility for all response decisions	Not responsible for decisions made on suggestions	Not yet formally established
Error of mapping = error of the organization	Post-quantum crypto: non-negotiable requirement	Gap documented as v0.x limitation

Principle of Non-Transfer

Responsibility does not transfer between levels. The framework defines the physics — universal and non-negotiable. The software suggests starting points — never endpoints. The organization decides the calibration and assumes full responsibility for the outcomes.

12 Computational Resilience

12.1 Hardware-Agnostic Architecture

The UBAID Framework was deliberately designed using gravitational vector field mathematics for a strategic reason: this mathematical language is hardware-agnostic. The same calculations that run today on classical computing run tomorrow on quantum computing, exponentially faster and with greater precision. The model does not need to be reformulated — it needs to be recompiled.

12.2 Native Quantum Analogies

Superposition	Entanglement	Tunneling
The Catman exists in multiple orbital states simultaneously until observation	Multiple simultaneous Astronauts and Entities with interdependent interactions	The Expectant Traveler Entity traverses the space indifferent to gravitational forces
The event collapses the state — precisely the quantum mechanical description	Effects on all agents can be calculated in parallel	Functionally analogous to quantum tunneling through barriers
Fluidity between legal and illegal orbitals is natively quantum	Ecosystem-scale simulation becomes computationally tractable	Provides formal physical model for Entity traversal behavior

12.3 Required Adaptations

- Theoretical model: no adaptation required — hardware-agnostic by construction
- Calculation engine: recompilation for quantum architecture when available — logic unchanged
- Data cryptography: CRYSTALS-Kyber, CRYSTALS-Dilithium, FALCON (NIST-standardized) from software v0.1 — foundational requirement, not future feature
- Communication protocols: TLS and inter-organizational exchange migrate to post-quantum algorithms before quantum maturity

13 Limitations and Open Frontiers

13.1 Documented v0.x Limitations

- The illegal universe is treated as uninhabited for the purpose of Astronaut position calculations because the framework does not yet possess precise mapping of the resources held by the origin sources of attacks. This is a methodological position required to preserve calculation

- integrity in v0.x, not an ontological claim. As attack-origin resources become mappable through framework evolution and community research, the model will expand to incorporate active inhabitants in the illegal universe.
- Block D.1 — structured vector recognition indicators for automated Layer 1 classification — is a pending deliverable, developed in parallel with v0.3.
- The Black Hole category formally acknowledges the existence of illegal celestial bodies whose internal nature is not yet describable. This boundary is named, not concealed. Black Holes exert observable gravitational force; their classification awaits research.
- The UBAID Research Organization (Level 3) does not yet exist formally. The singularity generator is currently fed by existing public corpora. This gap is documented as a v0.x limitation to be addressed with community support.

13.2 Open Research Frontiers

- Interaction physics between species: how does an Entity modify the gravitational field experienced by an Astronaut occurrence?
- The conditions and timing of the illegal universe becoming inhabited, and the expanded physics required when the resources of attack origins become mappable
- Formal mathematical definition of the Entity Threshold — precise conditions for Autonomous AI → Entity classification
- Cross-organizational orbital position comparison methodologies preserving model integrity
- Classification of Black Holes — establishing what these observable but currently unknown celestial bodies represent in the framework

14	Future Work	
	Version / Deliverable	Status
	v0.1 — Foundational Taxonomy	Published · GitHub
	v0.2 — Observable Indicators	Compiled · Draft for community review
	Block D.1 — Vector Recognition (parallel)	In development
	v0.3 — Entity Threshold Detection	Next milestone
	v0.4 — Legal Framework Alignment	Planned — GDPR, ISO 27001, EU AI Act
	Software Specification	Planned — reference implementation
	v1.0 — Peer Review Submission	Target

14.1 Derived Publications (to be developed with community support)

The following publications are planned to be developed in collaboration with the research community, not in isolation by the author. Each addresses a domain that extends beyond the scope of this foundational whitepaper and benefits from specialized expertise:

- Purple Team as Institutional Delegate — UBAID Applied to Red/Blue Team Operations
- Transgovernmental Cooperation Model — UBAID as Policy Framework for Multi-Jurisdictional AI Incident Response
- The Involuntary Catman — A Study of Cognitive Autonomy Erosion and Its Implications for Responsibility Attribution

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Conclusion

The UBAID Framework is not a prediction of a future disruption. It is a description of a present state that existing instruments are inadequate to address. The cognitive symbiosis between humans and artificial intelligence has already produced a new class of agent — the Catman — for which no adequate taxonomy of responsibility existed before this framework.

The gravitational physics model provides a mathematically precise, physically intuitive, and computationally agnostic language for describing the space in which these agents operate. The taxonomic classification provides the vocabulary for naming what we observe. The observable indicator system provides the operational instruments for classifying and positioning agents in real time. The governance model provides the architecture for implementing these instruments with clear and non-transferable responsibility at each level.

The historical pattern is consistent: hybrid agents precede their frameworks by decades. The harm accumulated in that interval is proportional to the scale of the amplification. Artificial intelligence is the largest amplification of human agency in history. The UBAID Framework is a formal attempt to compress that interval.

Author's Note

This framework is published as a draft for community review. The author invites critique, extension, and collaboration from researchers across cybersecurity, philosophy, law, AI governance, and related disciplines. Correspondence and contributions: github.com/upereirag/ubaid-framework

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